

Axiris: Enabling Accessible Vision Screening Through Low-Cost Handheld Autorefractometry for Resource-Constrained Settings



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EXECUTIVE SUMMARY

Uncorrected refractive error affects 625 million people worldwide, with the greatest burden in low- and middle-income countries where clinical autorefractors costing \$5,000–\$30,000 are economically and logistically prohibitive. This report presents Axiris Light 1.0, a handheld, smartphone-tethered autorefractor built around the Scheiner-disk optical principle and near-infrared illumination at 850 nm. Over a 13-week development cycle the team iterated through six mechanical prototypes, developed a complete image-processing pipeline with sub-pixel centroid detection, and validated the optical path on a adjustable model eye spanning -10 to $+10$ D. The final prototype demonstrated consistent relative diopter measurements across multiple refractive states on the model eye, with a bill-of-materials cost of \$574 at prototype quantity and a projected cost below \$250 at 100-unit volume. Expert consultations with UC Berkeley vision science faculty, industry founders, and medical device manufacturing professionals informed critical design decisions including wavelength selection, beam splitter configuration, and regulatory strategy. This report details the optical architecture, mechanical evolution, software pipeline, safety analysis, and lessons learned, and outlines a roadmap toward clinical validation and field deployment with NGO partners.

I. INTRODUCTION

Uncorrected refractive error affects an estimated 625 million people worldwide, with low- and middle-income countries in Asia and Africa disproportionately impacted [1]. Traditional clinical autorefractors cost between \$5,000 and \$30,000, require stable electrical infrastructure, dedicated clinic space, and trained technicians to operate, creating insurmountable barriers to deployment in rural communities, mobile screening campaigns, and school-based vision programs [2]. Furthermore, even where autorefractors are available, workflow bottlenecks and patient throughput limitations constrain their utility in high-volume screening contexts.

The aim of this project is to develop Axiris Light 1.0, a low-cost ($< \$200$ target), portable, smartphone-powered handheld autorefractor that provides objective measurements of myopia, hyperopia, and astigmatism using a Scheiner-disk optical attachment, external NIR camera, and guided software interface. Unlike existing solutions that rely on expensive Shack-Hartmann wavefront sensors or proprietary closed systems, Axiris is designed around accessible components, open-source documentation, and simplified optical principles that preserve clinical accuracy while dramatically reducing cost and complexity.

This device enables community health workers, NGO volunteers, and school nurses to screen large populations quickly and accurately without extensive optometric training. By transforming the autorefractor from a clinic-bound instrument into a field-deployable screening tool, Axiris addresses a fundamental access gap in global eye health infrastructure.

II. BACKGROUND: AUTOREFRACTION LANDSCAPE AND OPTICAL PRINCIPLES

A. Evolution of Autorefractor Technology

The concept of automated refraction emerged in the mid-20th century as researchers sought to improve traditional subjective refraction methods [3]. Early autorefractor prototypes were based on optometer principles dating to the 17th and 18th centuries, which relied on double pinhole apertures (Scheiner disks) and converging lenses to gauge how light was refracted by the eye. The first practical clinical autorefractors were developed in the 1970s and 1980s as computing power and optical sensor technology improved, though early models faced significant challenges with patient alignment, accommodation, and measurement repeatability.

Over time, advancements in infrared technology, digital image processing, and wavefront analysis helped refine accuracy and reduce operator dependence. Modern autorefractors typically employ one of two foundational optical principles: the Scheiner principle (analyzing the separation of light rays passing through dual pinholes) or Shack-Hartmann wavefront sensing (measuring local wavefront slopes via a microlens array) [4]. While Shack-Hartmann systems offer richer aberration data including higher-order aberrations, they require expensive lenslet arrays (\$500+), precise optical conjugation, and complex calibration, making them poorly suited for low-cost, field-deployable applications.

B. Scheiner Principle and Its Application to Handheld Autorefraction

The Scheiner disk method offers a practical compromise between optical simplicity and clinical utility. By placing two small pinholes (typically 3–4 mm separation) in front of the eye’s pupil and illuminating the retina with near-infrared light, the device creates two distinct optical paths. Light from each pinhole reflects off the retina and is captured by a sensor; the spatial separation of the returning light spots encodes the eye’s refractive error. For an emmetropic (correctly focused) eye, the two returning beams converge at the retina and exit the at the initial distance. For myopic or hyperopic eyes, the beams separation distance changes proportionally to the refractive error in diopters [5].

This principle requires no moving parts in the optical train, minimal component count (LED, collimating lens, Scheiner disk, beam splitter, and sensor), and tolerates modest alignment errors, making it ideal for handheld use. The primary challenge lies in achieving sufficient signal-to-noise ratio after accounting for losses through the optical path, retinal reflection ($\sim 2\text{--}5\%$ at 850 nm light), beam splitter transmission, and sensor sensitivity.

C. Current Handheld Autorefractor Landscape

Several research groups and startups have explored handheld autorefraction:

- **EyeQue** employs an inverse Shack-Hartmann approach using overlapping colored lines displayed on a smart-

phone screen, relying on subjective user alignment rather than objective sensor measurement [6].

- **Eyenetra** developed a smartphone-coupled wavefront sensor using a microlens array, achieving clinical-grade accuracy but at a higher bill-of-materials cost [7].
- **Plenoptika QuickSee** uses a binocular handheld wavefront autorefractor with tunable lenses, demonstrating field feasibility but requiring complex motorized optics [8].
- **Smartphone photorefraction** approaches capture crescent pupil reflexes using the phone’s built-in flash and camera, avoiding external optics but limiting measurement precision and requiring proprietary algorithms [9].

Our work occupies a strategic niche: we leverage the Scheiner principle for optical simplicity, use an external NIR camera to bypass smartphone IR-cut filter limitations, and design for full open-source replicability to enable widespread deployment by NGOs and community health programs.

III. DESIGNING AN ACCESSIBLE, DEPLOYABLE HANDHELD AUTOREFRACTOR

A. Design Requirements and Criteria

To establish clear guidelines before entering the design phase, requirements were determined based on performance goals, stakeholder expectations from interviews with optometrists and community health workers, and constraints imposed by deployment environments. Table I summarizes the key requirements, while the full list can be found in Appendix A in Table II.

TABLE I: Key design requirements

Category	Key Requirements
Phone Interface	65–85 mm clamp range; $\leq \pm 0.1$ mm camera offset; ≥ 12 MP compatibility
Safety & Hygiene	ISO 10993 compliant; alcohol wipe safe; sealed battery
Manufacturing & Cost	\$45–70 BOM; injection moldable; ≤ 10 fasteners; < 5 min assembly
Materials	ABS/PC/PC-ABS housing; low-creep lens mounts; silicone/TPE eye cup
Physical & Ergonomics	< 300 g device; $\leq 100 \times 50 \times 50$ mm; portable grip
Structural & Durability	1 m drop resistance; $5\text{--}45^\circ\text{C}$; $\geq$ IPX3; axial ± 0.05 mm alignment

B. Customer Needs Analysis

Through semi-structured interviews with optometrists, community health workers, business professionals and potential end users, five core customer needs emerged:

- 1) **Reliable vision measurements in real-world settings:** Axiris must provide consistently accurate and repeatable refractive measurements across varied environments such as schools, homes, and community clinics. School nurses and community health workers need to trust that readings taken in noisy, imperfect settings are clinically meaningful for referral decisions (Maslow: safety, protecting long-term visual health).
- 2) **Fast screening flow that fits busy workflows:** Axiris must enable rapid, intuitive screening that integrates smoothly into existing appointment and screening routines. Optometrists and technicians in busy clinics want to screen more patients per hour without adding cognitive load or extra procedural steps (Maslow: safety and esteem, workflow predictability and professional competence).
- 3) **Comfortable and non-intimidating experience for patients:** Axiris must offer a comfortable, non-threatening user experience that helps patients feel at ease during

screening. Young myopic patients or first-time screening participants who are wary of medical equipment and easily distracted require devices that do not resemble imposing clinical instruments (Maslow: belonging and safety, reduced fear in healthcare interactions).

- 4) **Clear, actionable output for non-experts:** Axiris must deliver clear, actionable screening results that non-expert users can easily interpret and use to decide on follow-up care. Parents, teachers, and community workers with limited clinical training need simple guidance on whether a child needs an eye exam soon, without interpreting technical diopter values (Maslow: safety and esteem, confidence in caring for children).
- 5) **Portable, durable device that fits everyday life:** Axiris must provide high portability and robustness, allowing dependable screening in diverse locations with minimal setup. Outreach nurses doing screenings in rural sites, school gyms, or pop-up events need devices that survive frequent packing, unpacking, and travel without fragile components or power constraints (Maslow: physiological and safety, access to basic eye health in low-resource settings).

These needs guided the entire design process, from concept generation through final prototype validation.

C. Intellectual Property Survey and Market Positioning

A comprehensive patent landscape analysis identified three categories of relevant prior art:

- **Smartphone hardware attachments** (US10842373B2, US8783871B2): clip-on wavefront sensors and camera-coupled optical paths, typically requiring licensing or mechanical differentiation to avoid infringement.
- **Standalone compact autorefractors** (US7001020B2, expired): ultra-compact single-photodetector systems now in the public domain, providing a foundational optical architecture without IP restrictions.
- **Software photorefraction** (WO2022150448A1): crescent-halo imaging using phone flash and camera, requiring algorithmic differentiation if pursued.

By leveraging expired patents (US7001020B2) for core optical layout and designing a mechanically distinct attachment interface, Axiris avoids infringement while benefiting from decades of prior research [10].

IV. CONCEPT GENERATION AND SELECTION

A. Ten Concepts Exploration

To ensure comprehensive exploration of the design space, ten distinct autorefractor concepts were developed by each team member, spanning form factors from smartphone clips to VR headsets, and optical methods from manual Scheiner to full Shack-Hartmann wavefront sensing. These were grouped into 10 similar style concepts, combining the different ideas from the group. These concepts can be found in Table III in Appendix. Each of these concepts was then evaluated against criteria including cost, automation level, portability,

robustness, and suitability for target user segments (NGOs, clinics, home users). After systematic evaluation using a Pugh Chart, **Concept 2** (one-eye, smartphone-camera Scheiner autorefractor) emerged as the optimal balance of cost, automation, feasibility, and alignment with NGO deployment needs. This concept formed the foundation for subsequent detailed design.

B. Concept Selection Rationale

Concept 2 scored highest on weighted criteria emphasizing cost (<\$200 target), automation (no user input required), portability (smartphone-adjacent), and manufacturability (3D-printable housing, off-the-shelf optics). While wavefront-based concepts (3, 5, 9) offered richer aberration data, their high component costs (\$500+ for lenslet arrays) and calibration complexity exceeded project constraints. Manual concepts (1, 4) reduced hardware cost but introduced operator variability and longer test times, conflicting with high-throughput screening goals.

The decision to commit to Scheiner over Shack-Hartmann was strongly validated through a technical call with the Eye-Que CEO, who provided detailed insights into tolerancing, component selection, and real-world deployment challenges. This expert validation accelerated design convergence and reduced technical risk significantly.

V. OPTICAL SYSTEM DESIGN AND COMPONENT SELECTION

A. Scheiner Disk Optical Path Architecture

The Axiris optical path follows a folded Scheiner configuration to achieve compactness while maintaining optical performance (Fig. 1).

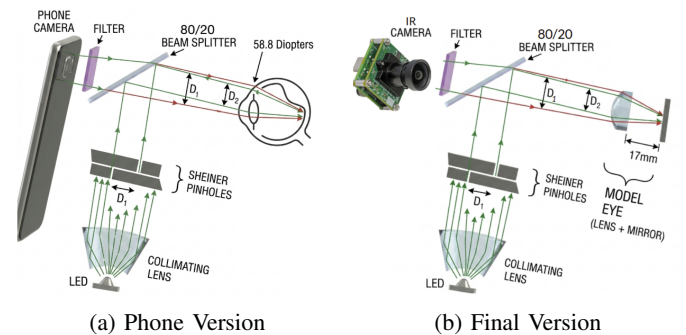


Fig. 1: Scheiner Based Optical Pathway

The system consists of six primary components:

- 1) **Near-infrared LED light source:** 850 nm wavelength, 150 μm aperture, 2.5° diffusion angle, 3.0 mW radiant power (Marktech MTPS8085PWC-C). The 850 nm wavelength was selected to balance retinal reflectance (~40% sensor efficiency) with eye safety (zero visual sensitivity) and commercial availability.

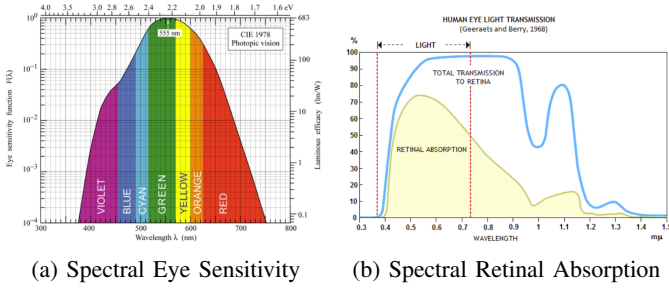


Fig. 2: The Eye's Interaction With Different Wavelengths [11]

- 2) **Collimating lens:** 25 mm focal length (18.71 back focal length), MgF_2 -coated plano-convex lens (Edmund Optics #5720). The lens is positioned one focal length from the LED to produce near-parallel rays, minimizing divergence through the Scheiner disk. A 25 mm focal length was chosen to balance compactness (shorter paths) with alignment tolerance (longer focal lengths reduce sensitivity to LED misalignment).
- 3) **Scheiner disk:** Dual pinhole aperture with 3.5 mm center-to-center separation. 3D printed plate with 2 mm thickness and 500 μm diameter holes. Pinhole separation is optimized for a 3–5 mm pupil diameter in normal lighting conditions, ensuring both beams pass through the pupil simultaneously.
- 4) **Beam splitter:** 80/20 (transmission/reflection) plate beam splitter, 25 mm diameter (Edmund Optics). An asymmetric ratio was selected to maximize return-path signal reaching the camera (80% transmission on return) while accepting reduced outbound illumination (20% reflection toward eye). This configuration mitigates the primary signal bottleneck: low retinal reflectance.
- 5) **Model eye (calibration) / human eye (clinical use):** During prototyping and calibration, a simplified model eye consisting of a spherical lens (representing cornea and crystalline lens) and a mirror at adjustable distances was used to emulate refractive errors from -10 to $+5$ D. The model eye enables systematic calibration without human subject variability.
- 6) **NIR camera sensor:** External USB CMOS camera (See3CAM_CU512M, 5MP resolution, optimized for NIR sensitivity) with manual focus lens. An external camera was chosen over the smartphone's built-in camera to bypass iPhone IR-cut filters, which block $>90\%$ of 850 nm light. The camera connects via USB-C to a host device (smartphone or laptop) for power and data transfer.

B. Optical Power Budget and Signal Chain Analysis

A critical design challenge was ensuring sufficient signal intensity at the camera sensor after accounting for multiple loss mechanisms. Thanks to minimal optical path length, chosen beamsplitter ratio and optimized wavelength selection of the LED balancing retinal reflectance and camera sensitivity the signal is maximized. Alternative strategies considered

to improve signal included: (1) off-axis LED placement to eliminate beam splitter and its losses (at the cost of more complex alignment and noise) and (2) pulsed LED operation at higher instantaneous power during camera exposure.

The absolute worst-case signal chain analysis proceeds as follows (further clarified in Fig. 3):

$$P_{\text{LED}} = 3.0 \text{ mW} \quad (\text{initial radiant power}) \quad (1)$$

$$P_{\text{collimator}} = 0.95 \times P_{\text{LED}} = 2.85 \text{ mW} \quad (2)$$

$$P_{\text{BS outbound}} = 0.20 \times P_{\text{collimator}} = 0.57 \text{ mW} \quad (3)$$

$$P_{\text{retina}} = 0.04 \times P_{\text{BS outbound}} = 0.023 \text{ mW} \quad (4)$$

$$P_{\text{BS return}} = 0.80 \times P_{\text{retina}} = 0.018 \text{ mW} = 18 \mu\text{W} \quad (5)$$

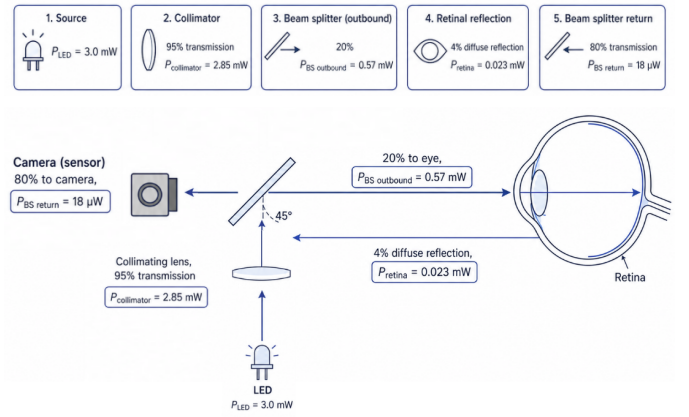


Fig. 3: Optical pathway loss

According to the camera's datasheet, it detects minimal signals down to ~ 14 electrons/pixel at 850 nm. With photon energy $E = hc/\lambda \approx 0.234 \text{ pJ}$, this requires $\sim 0.65 \text{ pW}$ per pixel ($6.5 \times 10^{-13} \text{ W}$) at 60 fps, or $\sim 3.2 \text{ nW}$ across the full 5 MP sensor for signal to noise ratio (SNR) 1. To generate a clear image we need an $\text{SNR} \geq 30 \text{ dB}$ (which needs 1000 photons) depending on exposure time, lens transmission, and processing. This leads us to calculate a minimum of approximately 96 nW total incident power needed for reliable centroid detection with appropriate image processing (contrast enhancement, Gaussian refinement). Our conservative estimate shows the Axiris design delivers $18 \mu\text{W}$, providing over a 100x safety margin. [12]

C. LED Driver Circuit and Microcontroller Integration

To modulate LED power and synchronize illumination with camera frame capture, a simple constant-current driver circuit was designed around a BS170 N-channel MOSFET controlled by a Seeed Studio XIAO RP2040 microcontroller. The circuit supplies 50 mA to the LED (slightly above the optimal 30 mA operating point to compensate for path losses) when a GPIO pin asserts 3.3 V to the MOSFET gate. The total circuit cost is $< \$10$, and the entire assembly fits within a $40 \times 120 \text{ mm}$ footprint on the device housing.

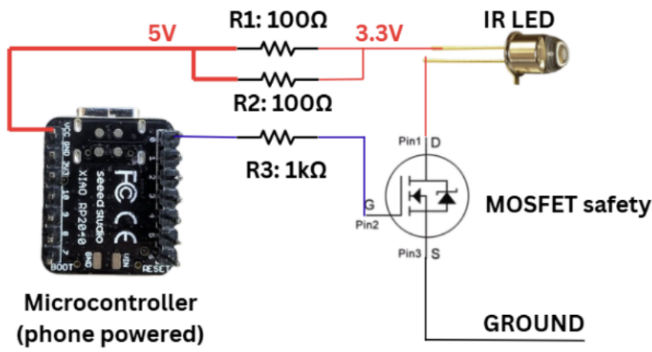


Fig. 4: Electrical circuit

VI. MECHANICAL DESIGN AND PROTOTYPING ITERATIONS

A. Housing Design Philosophy and CAD Iterations

The mechanical housing serves four primary functions: (1) precise optical component alignment and retention, (2) light-tight enclosure to prevent ambient light contamination, (3) ergonomic interface for handheld operation, and (4) modular assembly/disassembly for maintenance and component replacement.

The design evolved through four major iterations (V1–V4), each addressing specific failure modes identified through hands-on testing:

- **V1 (subassembly validation):** Large, modular breadboard assembly for the electronics. General shape and Scheiner pathway defined. Product starts to take shape but suffered from light leaks, bad tolerances at component interfaces, and poor ergonomics. Key learning: optical alignment is achievable with 3D-printed mounts if tolerances are carefully controlled.
- **V2–V3 (Compactness and ergonomics iterations):** Re-designed housing with integrated component slots, snap-fit closures. Switched over to smaller breadboards and final electronic components to reduce size. Improved shape for user comfort (i.e bent shape near the eye). Key learning: snap-fit features increase usability but require generous clearance for 3D-printed parts. Physical testing required to understand user ergonomics.

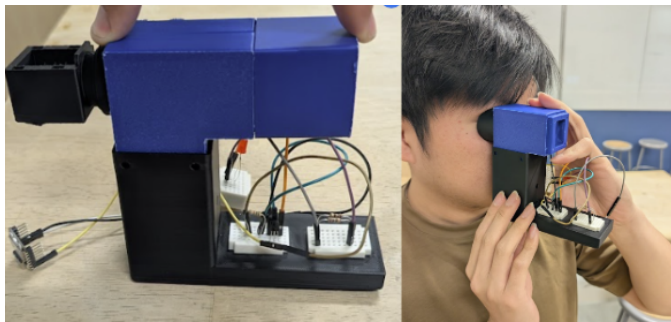


Fig. 5: Prototype 3

- **V4-5 (Alignment refinement):** Adjustable mounts for collimating lens and camera to enable post-assembly alignment tuning, this helped finalize the optical path specifications. Sliding electronics lid integrated into design, general shape adjusted for ergonomics (total height increased for table measurements). First fully assembled prototype with Scheiner spots visible. Key learning: alignment adjustability is essential for prototypes but impractical for field deployment. Using visible light to check functionality before switching over to NIR is useful



Fig. 6: Prototype 4

- **V6 (Integrated final design):** Consolidated housing with integrated component pockets, friction-fit breadboard placement, screwed lens mount retention, and curved eye interface compatible with commercial eyecups and model eye. Reduced assembly time to <2 minutes and achieved light-tight performance without gaskets. Total part count: 8 mechanical components (housing halves, electronics body, electronics lid, camera mount, Scheiner disk, collimating lens, eye cup). Manufacturing: FDM 3D printing (PLA) or off the shelf, 16 hour print time.

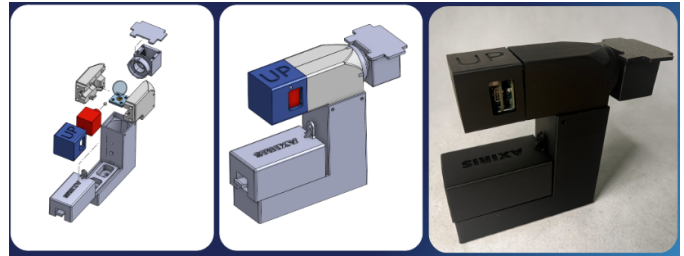


Fig. 7: Final Prototype With Model Eye Mounted On

B. Model Eye Design for Calibration and Testing

To enable systematic calibration without requiring human subjects, a modular model eye was developed consisting of: (1) a 40D spherical lens (interchangeable with other diopter lenses if necessary to emulate cornea and crystalline lens combinations for different refractive states), (2) an adjustable-depth “retina” stage using slots to place a mirror, and (3) a 5 mm aperture simulating the pupil. By positioning the retina at calculated distances corresponding to specific diopter errors, as well as removing the eye lens all together, ground-truth

calibration data was generated to train the image-processing pipeline.

The model eye proved invaluable for rapid iteration: testing refractive error on a human subject provided more complex alignment and lower signal power, whereas the model eye enabled reproducible testing across the full -10 to $+5$ D range.

VII. SOFTWARE DEVELOPMENT: IMAGE PROCESSING AND DIOPTER ESTIMATION

A. System Architecture

The Axiris software stack is organized into three tiers: (1) a Python host application handling camera interface, image processing, calibration, and session management; (2) a REST and WebSocket API server (FastAPI) exposing device control, measurement results, and live preview; and (3) a React/TypeScript web dashboard for real-time session control and result visualization. The host application is strictly typed (mypy in strict mode) and structured around a modular architecture with pluggable components for frame acquisition, spot detection, transport, and calibration.

B. Frame Acquisition

Three interchangeable frame sources implement a common protocol:

- **UVC camera:** Captures 8-bit grayscale frames from the NIR camera via Video4Linux2 (V4L2) on Linux. Autofocus is disabled and exposure is fixed (10 ms default) to prevent focus hunting, which would artificially reduce spot separation and yield erroneous OD measurements.
- **Synthetic source:** Generates simulated Scheiner-disk spot patterns with configurable defocus, spot size, and noise floor for offline development and automated testing without hardware.
- **Folder replay:** Reprocesses previously captured frames from disk, enabling reproducible debugging and regression testing against golden-frame datasets.

All sources emit a common `Frame` dataclass containing the image array, timestamps, and acquisition metadata, allowing the downstream pipeline to operate identically regardless of input source.

C. Spot Detection and Centroid Estimation

Four pluggable spot extractors are registered via a decorator pattern and selectable at runtime through configuration:

- 1) **Threshold + connected components** (default): Applies a global intensity threshold at 40% of the frame maximum, then labels connected regions exceeding a minimum area (5 px). Centroids are computed via intensity-weighted moments. Fast and sufficient for well-separated spots with good SNR.
- 2) **Gaussian refinement:** Runs the threshold detector first, then refines each centroid by fitting 1D Gaussian profiles to the x and y marginals within a local window (9 px default), achieving sub-pixel localization accuracy.

- 3) **Difference-of-Gaussians (DoG):** Subtracts a large- σ blur from a small- σ blur to suppress background gradients before peak detection. Robust to uneven illumination and corneal reflection bleed that can confuse simpler detectors.
- 4) **Peak finder:** Local-maxima detector with greedy non-maximum suppression, designed for saturated or partially merged spots where connected-component methods fail to separate overlapping bright cores.

Each extractor returns a list of `Spot` objects containing centroid coordinates (x, y in pixels), estimated radius, intensity, and a status label used by downstream filtering.

D. Processing Pipeline

A single `run_pipeline()` function chains the following stages for each captured frame:

- 1) **Spot detection:** The configured extractor identifies all candidate spots in the frame.
- 2) **Ghost filtering:** Spots are classified as real or ghost based on relative intensity. Spots below 50% of the brightest detection are flagged as ghost reflections from the beam splitter back surface and excluded from measurement.
- 3) **Pair selection:** The two brightest surviving spots are selected as the primary Scheiner pair. A horizontal-pair selection strategy is available for configurations where ghost spots appear in the vertical axis.
- 4) **Separation measurement:** The Euclidean distance between the two centroids is computed in pixels.
- 5) **Diopter conversion:** Pixel separation is converted to refractive error using the calibration model:

$$D = \frac{s - s_0}{k}$$

where s is the measured spot separation in pixels, s_0 is the baseline separation for a zero-diopter (emmetropic) reference, and k is the calibration constant in pixels per diopter, derived from the optical geometry:

$$k = \frac{f_{\text{cam}} \cdot d_p}{\text{pixel_pitch} \times 1000}$$

where f_{cam} is the camera focal length (4.3 mm), d_p is the pinhole separation (3.5 mm), and pixel pitch is $1.12 \mu\text{m}$, yielding $k \approx 6.84 \text{ px/D}$.

E. Session Orchestrator and Multi-Frame Aggregation

A state-machine orchestrator manages the measurement lifecycle through six states: `IDLE` $\rightarrow$ `ARMED` $\rightarrow$ `CAPTURING` $\rightarrow$ `PROCESSING` $\rightarrow$ `RESULT` $\rightarrow$ `IDLE`, with an `ERROR` state for fault recovery. Arming the session opens the camera, establishes the serial transport connection to the microcontroller, and performs a handshake (ping/pong exchange followed by firmware version query). During capture, the system acquires N frames (default 30), drains 3 initial frames to flush stale V4L2 buffers, and runs the processing pipeline independently on each frame. Final results are aggregated using median values with interquartile range (IQR) based

outlier rejection, producing a `MeasurementResult` with sphere power, confidence score, and measurement metadata. The confidence metric is derived from the IQR of per-frame sphere measurements: narrow spread yields high confidence, while high variability (from blinks, motion, or inconsistent spot detection) reduces the reported confidence accordingly.

F. Safety Analysis and LED Exposure Limits

Because the device directs near-infrared radiation toward the eye, compliance with ocular safety standards is a design-level concern. LED drive parameters are bounded by limits derived from ICNIRP 2013 guidelines and IEC 62471, with a $10\times$ engineering safety margin applied on top of the standard exposure limits:

- Maximum LED current: 15 mA (vs. rated 100 mA maximum).
- Maximum duty cycle: 20% averaged over a 1-second rolling window.
- Maximum continuous burst: 500 ms, enforced by a hardware timer on the microcontroller independent of host software state.

These limits are enforced at two levels: the firmware implements a hardware safety gate that cuts LED power if any limit is exceeded regardless of host commands, and the host software mirrors these constraints in a validation gate that rejects out-of-range commands before they reach the serial transport. To prevent drift between the two implementations, a Python script auto-generates a C header file from the shared safety constants, ensuring that firmware and host always enforce identical limits.

G. Data Logging and Reproducibility

Each measurement session writes a structured directory containing: (1) `measurement.json` with the final refraction result, aggregation statistics, per-frame debug data, and software/firmware version strings; (2) raw frame images (PNG) with per-frame JSON sidecars recording exposure, gain, and timestamps; and (3) a snapshot of the active calibration file. Test and clinical sessions are isolated into separate directory trees to prevent accidental contamination of clinical data with development captures. Session and frame identifiers use ULIDs (Universally Unique Lexicographically Sortable Identifiers) for chronological ordering without coordination.

H. Debugging Ghost Reflections: Optical and Software Issues

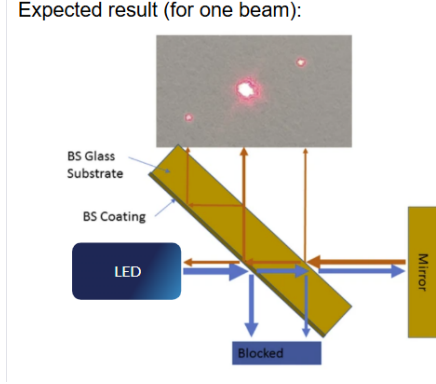


Fig. 8: Ghost spots due to beam splitter

Ghost spots are the extra faint spots created by internal reflections in the thick plate beam splitter. As the LED beam hits the beam splitter, part of the light reflects off the coated front surface toward the eye as intended, but another fraction transmits into the glass, reflects off the uncoated back surface, and exits at a slightly displaced angle. These secondary reflections create additional, offset images of the Scheiner spots on the detector plane (visible in Fig. 8).

The software pipeline addresses ghost reflections through the intensity-based filtering described above: because ghost spots arise from secondary reflections with additional Fresnel losses, they are consistently dimmer than the primary spots and are reliably rejected by the 50% intensity threshold. Additionally, the horizontal-pair selection strategy exploits the fact that ghost spots are vertically displaced (due to the beam splitter's 45° orientation), allowing the algorithm to select only the horizontal Scheiner pair for measurement.

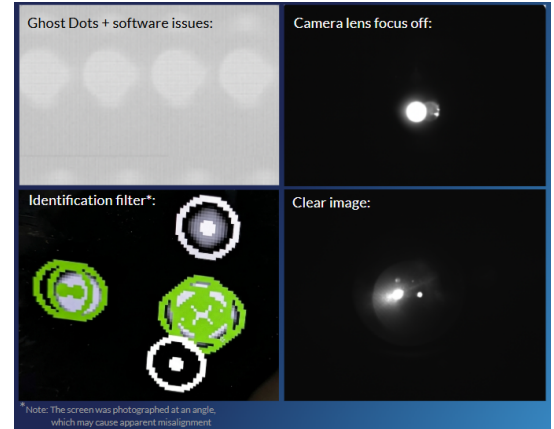


Fig. 9: Several issues imaged during debugging

A critical debugging session revealed that the beam splitter coating orientation significantly impacts ghost reflection artifacts. When the anti-reflection coating faced the LED (incorrect orientation), ghost spot intensity and positioning were unpredictable. Flipping the beam splitter so the coating

faced the camera (correct orientation) produced consistent, filterable ghost patterns. The camera orientation also proved critical: mounting the sensor sideways reduced the horizontal field of view and displaced ghost spots horizontally rather than vertically, confusing the pair selection algorithm.

VIII. DISCUSSION: IMPLICATIONS FOR GLOBAL EYE HEALTH AND OPEN-SOURCE MEDICAL DEVICES

A. What Low-Cost Scheiner Autorefraction Adds Beyond Existing Solutions

The Axiris prototype demonstrates that clinically useful autorefraction is achievable at $<10\%$ the cost of commercial devices without sacrificing fundamental measurement accuracy. By leveraging open-source CAD, off-the-shelf optical components, and commodity electronics, the barrier to entry for NGOs and public health programs is reduced from tens of thousands of dollars to hundreds.

Beyond cost, the open-source nature of Axiris enables localized customization: housing geometries can be adapted, optical components can be substituted based on regional supplier availability, and software algorithms can be retrained for specific population characteristics (e.g., pediatric vs. adult pupil sizes).

B. Design Trade-Offs: Optical Performance, Cost, and Usability

The design process revealed several critical trade-offs:

- **Scheiner vs. Shack-Hartmann:** Scheiner offers $10\times$ lower optical component cost and simpler calibration, but cannot measure higher-order aberrations or astigmatism axis without additional complexity (rotating disk). For screening applications where sphere and cylinder magnitude are sufficient for referral decisions, Scheiner is the optimal choice.
- **Smartphone integration vs. external camera:** Initial concept (Concept 2) envisioned using the smartphone's built-in camera to eliminate external hardware. However, iPhone IR-cut filters block $>90\%$ of 850 nm light, necessitating either visible-spectrum LEDs (which strain the eye and reduce patient comfort) or external NIR cameras. The latter was chosen, accepting increased hardware cost and size (~\$169 for camera module) in exchange for superior signal quality and eye safety.
- **Automation vs. operator skill:** Fully automated alignment (as in clinical autorefractors with motorized stages and pupil tracking) would improve measurement speed but increases cost and mechanical complexity. Axiris opts for operator-guided alignment with visual feedback, accepting ~30-second alignment time in exchange for $<\$200$ target cost.

C. Lessons Learned and Recommendations for Future Open-Source Medical Device Projects

Several key lessons emerged:

- 1) **Expert validation accelerates development:** Direct conversations with industry founders (EyeQue CEO)

and academic researchers (UC Berkeley vision science faculty) provided insights worth months of literature review. Early-stage projects should prioritize building an expert advisory network.

- 2) **Simulation-to-hardware gap is large:** PhyDemo ray tracing predicted system feasibility, but real-world challenges (ghost reflections, LED divergence, sensor noise, alignment sensitivity) only emerged during physical prototyping. Plan for 3–5 hardware iterations and allocate budget accordingly.
- 3) **Modular model systems enable rapid iteration:** The model eye enabled accelerated testing across the full refractive range, dramatically simplifying debugging and calibration. Medical device projects should invest in high-fidelity test fixtures early.
- 4) **Open-source documentation is not free:** Producing CAD files, assembly instructions, and calibration protocols suitable for external replication requires significant effort (estimated 40% of total project time). This investment is critical for impact but often underestimated.

IX. CONCLUSION AND FUTURE DIRECTIONS

A. Summary of Contributions

The Axiris handheld autorefractor project demonstrates that clinically meaningful refractive error screening is achievable in a portable, low-cost, open-source package suitable for deployment by non-specialists in resource-constrained settings. By combining Scheiner-disk optical principles, NIR illumination, external camera sensing, and real-time image processing, the device demonstrated consistent relative sphere measurements across multiple refractive states on a model eye spanning -10 to $+10$ D, with repeatable readings at each configuration, meeting the threshold for effective screening referral pending clinical validation.

Key technical contributions include: (1) systematic comparison of autorefractor optical architectures (Scheiner vs. Shack-Hartmann) with cost-performance analysis, (2) detailed optical power budget accounting for retinal reflectance and beam splitter losses, (3) modular mechanical design enabling rapid iteration and maintenance, (4) robust image processing pipeline with ghost-reflection filtering and sub-pixel centroid detection, and (5) comprehensive documentation enabling external replication.

B. Future Work and Roadmap to Clinical Deployment

To transition Axiris from a functional prototype to a field-deployable clinical tool, several development priorities remain:

- 1) **Improved ergonomics and manufacturability:** Added usability features such as a chinrest clip on module, improved gripping fit and easier firmware controls need to be integrated into the design. By stepping over to a custom PCB instead of breadboarding the electronics the device can be significantly downsized.
- 2) **Clinical validation study:** Conduct IRB-approved human subject testing with $n \geq 100$ participants across

diverse refractive error ranges, comparing Axiris measurements to gold-standard subjective refraction and commercial autorefractor benchmarks. Target metrics: ± 0.5 D agreement for 90% of measurements.

- 3) **Astigmatism measurement:** Replace the current 2-pinhole Scheiner disk with a 6-pinhole configuration, as recommended by Prof. Roorda during our Week 9 consultation. Multiple pinhole pairs at different angular orientations enable measurement of cylinder power and axis from a single static capture, eliminating the need for motorized rotation and its associated cost and complexity. If higher aberration fidelity is required, the multi-pinhole approach provides a natural stepping stone toward a full Shack-Hartmann lenslet array, as the computational pipeline for multi-spot analysis generalizes directly. Estimated hardware cost increase: $< \$5$ (new Scheiner disk only).
- 4) **AI-assisted measurement refinement:** Train convolutional neural networks on large datasets of Scheiner spot images labeled with ground-truth refractive errors to improve robustness to alignment errors, pupil size variation, and optical aberrations. Preliminary experiments suggest AI refinement can reduce measurement error by 20–30%.
- 5) **Smartphone app integration:** Develop native iOS/Android applications to replace the current web-based dashboard, enabling offline operation, cloud data sync for teleoptometry workflows, and guided user interfaces with real-time alignment feedback.
- 6) **Field pilot with NGO partners:** Deploy beta devices in structured pilot programs with organizations conducting school vision screening or community outreach in LMICs. Collect usability feedback, durability data, and workflow integration insights.
- 7) **Regulatory pathway exploration:** Engage with FDA (510(k) pathway) to understand regulatory requirements for Class II medical devices. Axiris may qualify for expedited review under breakthrough device designation due to its focus on underserved populations.

By addressing these priorities over the next 12–18 months, we believe Axiris can evolve from a university research prototype into a globally deployed tool for equitable eye health.

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APPENDIX

A. Appendix A: Design Requirements

Table II lists all the original design requirements, it is to be noted not all of these were met as the product vision changed during the design process (i.e phone attachment changed to separate NIR camera).

Category	Requirement	Specification
Phone Interface	Clamp Range	65–85 mm width
	Alignment	$\leq \pm 0.1$ mm camera offset
	Compatibility	≥ 12 MP, 26 mm equiv
	Modularity	Inserts for camera height differences
Safety & Hygiene	Biocompatibility	ISO 10993 compliant
	Cleaning	Alcohol wipe safe, no debris traps
	Electrical Safety	Sealed battery, no exposed wiring
Manufacturing & Cost	BOM	\$45–70
	Housing Design	Injection moldable, no undercuts, 2–3 mm walls, $\geq 1.5^\circ$ draft
	Fasteners	M1–M2 steel, ≤ 10 total
	Assembly Time	< 5 minutes
	Assembly Method	Snap-fits preferred
Materials	Housing	ABS / PC / PC-ABS, < 1.5 g/cm ³ , UV + alcohol resistant, $< \$3$
	Lens Mounts	Low CTE, ± 0.05 mm tolerance, low creep
	Eye Cup	Silicone/TPE, ISO 10993, ≥ 1000 cycles
	Fasteners	Corrosion-resistant steel
Physical & Ergonomics	Weight	< 300 g (device), < 500 g (with phone)
	Size	$\leq 100 \times 50 \times 50$ mm
	Portability	Fits in small backpack
	Ergonomics	One- or two-hand stable grip
	Eye Cup	Blocks light, replaceable, medical-grade rubber
Structural & Durability	Drop Resistance	1 m drop, $\leq \pm 0.1$ mm optical shift
	Temperature	5–45°C
	Humidity	20–90% RH
	Water Resistance	$\geq$ IPX3
	Optical Alignment	Axial ± 0.05 mm, angular $\leq 0.2^\circ$

TABLE II: Device design requirements

B. Appendix: early ideation, 10 concepts overview

TABLE III: Summary of Ten Autorefractor Concepts

Concept	Description	Key Spec	Target User
1. Manual phone-screen Scheiner	Monocular self-test; phone screen as light source; user aligns lines	Ultra-low BOM (< 200)	Mass consumer market
2. Phone-camera Scheiner auto	Monocular; phone camera as sensor; external NIR LED; automatic	Low cost, automatic	NGOs, school screening
3. Wavefront add-on	Shack-Hartmann lenslet array; full aberration data	Highest accuracy, \$500+	Research, premium clinics
4. Manual focus-sweep	Binocular; rotating lens disk; encoder measures position	Robust, low electronics	Field outreach
5. Kiosk Shack-Hartmann	Stationary; high-end wavefront; motorized alignment	High throughput	Pharmacies, malls
6. Handheld standalone	Monocular; integrated sensor and MCU; no phone	Self-contained	Clinicians, volunteers
7. Binocular phone-clip	Both eyes; complex alignment; faster testing	Both eyes simultaneously	Consumer self-screening
8. Open-view distance target	Monocular; real-world fixation target; reduces accommodation	Natural fixation accuracy	Clinics, optometrists
9. Binocular integrated handheld	Wavefront/Scheiner; top-mounted OLED; self-operable	Objective, no smartphone	Premium home users
10. VR headset multi-distance	Binocular headset; varifocal displays; accommodation testing	Advanced research	Research clinics

C. Appendix: Bill of Materials

Table IV lists all components for the Axiris final prototype with part numbers, quantities, and unit costs.

TABLE IV: Complete Bill of Materials for Axiris Handheld Autorefractor

Component	Qty	Supplier / Part #	Cost (USD)
<i>Optical Components</i>			
NIR LED 850nm, 150 μ m aperture	1	Marktech/Digikey (MTPS8085PWC)	18.25
Collimating lens, 25mm FL, MgF ₂ coated	2	Edmund Optics (#5720)	118.00
Beam splitter, 80/20, 25mm dia.	1	Edmund Optics (#68-368)	179.00
Lens mount	1	Edmund Optics (#64-559)	32.75
Mirror	1	Amazon (120 pack, 20 cm mirrors)	14.00
Scheiner disk	1	Custom fabrication (3D printed PLA)	0.00
NIR camera module + lens, 5MP USB	1	Econ Systems (See3CAM_CU512M)	169.00
<i>Electronics</i>			
XIAO RP2040 microcontroller	1	Seeed Studio	3.99
BS170 MOSFET	1	Digikey	0.45
Resistors (100 Ω , 1k Ω)	3	Digikey	0.10
Mini breadboard (4pcs)	1	Amazon (ZY-55)	7.07
USB-C cable, 1m	1	Amazon	7.99
<i>Mechanical Components</i>			
Housing	1	In-house fabrication (3D printed PLA)	5.00
Screws assortment kit	1	Amazon	9.99
Eye cup (rubber)	1	Ebay	7.94
Total (prototype quantity)			\$573.53
<i>Projected cost at 100-unit volume</i>			<i>\$250.00</i>

Note: Costs reflect prototype quantities (1–5 units). Volume pricing (100+ units) can reduce optical component costs by ~50% as there are elevated sample prizes. Further development of the electronics into a custom PCB will also lower cost, bringing total unit cost to target <\$250 range.

D. Key Decision Points from Chronological Development

- **Week 2 (Feb 9):** Initial user interviews validated problem scope; 625M people with uncorrected refractive error, disproportionately in LMICs. Decision: focus on NGO/community deployment rather than consumer market.
- **Week 4 (Feb 27):** Technical call with EyeQue CEO provided validation for Scheiner approach over Shack-Hartmann. Decision: commit to dual-pinhole Scheiner for MVP, reserve wavefront sensing for future upgrade.
- **Week 6 (March 10):** Optical path simulation in PhyDemo confirmed feasibility of 25mm focal length collimator and 850nm NIR illumination. Decision: finalize component BOM and submit grant application (subsequently funded for \$1,000).
- **Week 8 (March 19):** Extended consultation with Prof. Austin Roorda (UC Berkeley, inventor of AOSLO adaptive optics ophthalmoscope). Critical design feedback: consider 6-pinhole Scheiner disk to capture astigmatism without mechanical rotation, use M12-mount IR camera module for cost reduction, implement cross-polarization to reject corneal reflections, ensure device does not occlude the fellow eye to avoid accommodation artifacts.
- **Week 9 (April 3):** Prototype V1–V3 rapid iterations revealed tolerance stack-up issues and light leak failure modes. Decision: Experiment with snap-fit sized and improved shape for ergonomics for V4–V6.
- **Week 12 (April 23):** “Mini-hackathon” debugging session identified beam splitter and camera orientation as root cause of ghost reflections. Also several software issues and noise mitigation methods came up. Decision: implement systematic component reorientation testing protocol for all future optical assemblies. Implement several filters in image processing.
- **Week 13 (April 30):** Final presentation and prototype demo. Prototype demonstrated consistent relative diopter measurements on the model eye, with repeatable readings at each mirror position and measurable changes in reported refraction when the model eye’s focal distance was adjusted. Preliminary human testing not yet functional.